

Siavash Mirarab (Mir arabbaygi)

CONTACT INFORMATION	UC San Diego (ECE) 9500 Gilman Drive, Mail code 0407 La Jolla, CA 92093-0407	e-mail: smirarab@ucsd.edu , phone: 858 822 6245 http://ecweb.ucsd.edu/~smirarab/ google scholar: http://goo.gl/gEswQ5
ACADEMIC POSITIONS	UC San Diego (UCSD) , San Diego, CA, United States Department of Electrical and Computer Engineering <i>Assistant Professor</i> July 2015 – July 2020 <i>Associate Professor</i> July 2020 – July 2025 <i>Full Professor</i> July 2025 – present	
EDUCATION		
<i>Ph.D</i>	University of Texas at Austin , US	Jan. 2011 – July 2015 Computer Science Department; Advisor: Prof. Tandy Warnow (& Keshav Pingali)
<i>M.S</i>	University of Waterloo , ON, Canada	Sept. 2006 – Sept. 2008 Electrical and Computer Engineering; Advisor: Prof. Ladan Tahvildari
<i>B.S</i>	University of Tehran , Tehran, Iran	Sept. 2001 – Dec. 2005 Electrical and Computer Engineering (Specialized in software engineering)
AWARDS	Clarivate Highly Cited Researcher for 2022 – 2025 2023 – 2026 Test of time award for REOMB 2014 (for our paper on PASTA) 2026 Best paper awards for SSB (Systematic Biology) and RECOMB-CG 2023 Best student award runnerup for MBE 2021 NIH-NIGMS MIRA award 2021 NSF CAREER award 2019 Alfred P. Sloan Research Fellow 2017 Honorable Mention for the 2015 ACM Doctoral Dissertation Award. 2016 Bert Kay Dissertation Award, UT, Austin, Computer Science 2016 Howard Hughes Medical Institute International Student Fellowship. 2012 - 2015 National Science and Engineering Research Council of Canada (NSERC), Canada Graduate Scholarship (CGS-D) converted to PGS-D outside Canada. 2011- 2012 Other awards: UT, Austin, College of Natural Sci., Dean’s Excellence Award (2011), Dean’s Prestigious Fellowship Award (2011 – 2014), and MCD fellowship (2011)	
RESEARCH FUNDING	NSF: DMS-2601546: “A mathematical framework for scalable species tree ...” 2026 Minderoo Found.: “Blue Microbes ...”. Co-PI. Share: \$930 K (2 years) 2025 Minderoo Found.: “Enabling the measurement of marine...” \$726 K (2 years) 2022 NIH: R35GM142725: “Biology-aware machine learning...” \$1.7 M (5 years) 2021 NSF: CISE-1845967: “CAREER: Robust & scalable genome...” \$549k (5 years) 2019 NSF: CISE-1815485 : “III: SMALL: New algorithms for genome...” \$500k (3 years) 2018 NSF: CISE-1565862: “CRII: III: Using genomic context to ...” \$170k (2 years) 2016 NIH-CFAR: “Accuracy of HIV transmission network ...” \$439K (2 years) 2016 CFAR, “Algorithms for prioritizing prevention intervention ... HIV”, \$50K 2019–2020	

	Sloan foundation fellowship, \$60K (2 years)	2016–2018
	Other gifts and internal grants ≈\$300K	2017–2024
WORK EXPERIENCE TEACHING:	<i>IBM, Vancouver, BC, Canada IT Specialist</i>	Sept. 2008 – Dec. 2010
	Course Instructor (University of California, San Diego):	
	Graduate: Computational evolutionary biology (ECE 208)	2016 – 2026
	Introduction to Computer Engineering (ECE 30)	2016 – 2026
HIGHLIGHTS AND INVITED TALKS	See http://eceweb.ucsd.edu/~smirarab/presentations.html for a complete list	
	Institute for Mathematical and Statistical Innovation (IMSI), Chicago, IL	2025
	Computational Genomics Summer Institute (CGSI), Los Angeles, CA	2017 – 2025
	Evolution Meeting, Athens, GA, Albuquerque, NM	2023, 2025
	Institute for Comp. and Exp. Research in Mathematics (ICERM), Rhode Island	2024
	SMBE, Puerto Vallarta, AI in genomics,	2024
	Mathematics of Evolution, Singapore,	2023
	RECOMB-seq, San Diego, keynote speaker,	2022
	Asilomar Conference on Signals, Systems, and Computers, Monterey,	2022
	OH-Know!, Norway (online),	2021
	Phyloseminars, online,	2021
	Grainger Bioinformatics Center Symposium, online,	2020
	SysBioDiv and Zoological Society, Freiburg, Germany	2020
	QCB Colloquium, USC, Biological Sciences, Los Angeles, CA	2019
	Workshop on Microbiomics, Metagenomics, and Metabolomics, Buffalo, NY	2019
	Genome 10K & Vertebrate Genome Project, New York, NY	2019
	Avian Meetings, Beijing, China, San Diego, CA (with PAG)	2016, 2018, 2019
	Evolution meeting, Austin, TX, Portland, OR, Montpellier, France	2016, 2017, 2018
	ENABLE workshop, National Chung Hsing University, Taiwan	2018
	Institute for Pure & Applied Math, Workshops on MSA and HPC	2015, 2018
	ISMB, Invited Tutorial on comparative regulatory genomics, Chicago, IL	2018
	Simons Found., Workshop on Statistical Challenges in Microbiome, Boston, MA	2017
	University of Copenhagen, Seminar Series, Copenhagen, Denmark	2017
	Schloss Dagstuhl – Leibniz Center for Informatics, 16351, Dagstuhl, Germany	2016
	RECOMB, Highlights session, Warsaw, Poland	2015
	INFORMS, Big Biological Data, Philadelphia, PA	2015
	MolPhy-3, Moscow, Russia	2012
PROFESSIONAL SERVICES	Leadership: Vertebrate Genome Project (VGP), Council member.	2020–present
	UCSD: Curriculum Development Committee. “Engineering in Public Health”.	2020
	Conference organization: (IEEE BIBM) (2019), REOMB (2021)	2019–2020
	Program Committee: ISBM, ECCB, RECOMB-CG, RECOMB	2015, 2017–2024
	Paper Review for journals of wide readership (e.g., Science, Nature methods), in biology (e.g., Nature Ecology and Evolution), in math (e.g., Journal of the American Statistical Association), and in computer science (Journal of ACM).	
	Grant Panels: NSF, DOE, NIH	2018–2023

1. Iker Rivas-González, [Siavash Mirarab](#) and Edward L. Braun. Phylogenomics beyond the ‘bag-of-genes’ model. *Trends in Genetics*, S0168952526001150 2026.
2. Eduardo Charvel, Isaac Thomas, Homère J. Alves Monteiro et al. ReSkmer: modeling repeats allows k-mer-based alignment-free methods to calculate population genomic distances. *Genome Biology*, Ahead of print 2026.
3. Yasamin Tabatabaee, Santiago Claramunt and [Siavash Mirarab](#). Coalescent-based branch length estimation improves dating of species trees. *Systematic Biology*, syag038 2026.
4. Daira Melendez, Ali Osman Berk Şapcı, Vineet Bafna and [Siavash Mirarab](#). SPrUCE: Utilizing Ultraconserved Elements of DNA for Population-Level Genetic Diversity Estimation. *Molecular Ecology Resources*, 26(3):e70145 2026.
5. Shayesteh Arasti, Puoya Tabaghi, Yasamin Tabatabaee, Mayer, Alan K. and [Siavash Mirarab](#). Optimal Tree Metric Matching Enables Phylogenomic Branch Length Reconciliation. *Systematic Biology*, syag025, 2026.
6. Ali Osman Berk Şapcı and [Siavash Mirarab](#). krepp: a k-mer-based maximum pseudo-likelihood method for estimating read distances and genome-wide phylogenetic placement. *Genome Biology*, 27(1):108 2026.
7. Yasamin Tabatabaee, Chao Zhang, Shayesteh Arasti and [Siavash Mirarab](#). Species Tree Branch Length Estimation despite Incomplete Lineage Sorting, Duplication, and Loss. *Genome Biology and Evolution*, 17(11):evaf200 2025.
8. Eleonora Rachtman, Yueyu Jiang and [Siavash Mirarab](#). Machine Learning Enables Alignment-Free Distance Calculation and Phylogenetic Placement Using k -Mer Frequencies. *Molecular Ecology Resources*, 25(8):e70055 2025.
9. Ramunas Stepanauskas, Julia M Brown, Shayesteh Arasti et al. Net rate of lateral gene transfer in marine prokaryoplankton. *The ISME Journal*, 19(1):wraf159 2025.
10. Chao Zhang, Rasmus Nielsen and [Siavash Mirarab](#). ASTER: A Package for Large-scale Phylogenomic Reconstructions. *Molecular Biology and Evolution*, msaf172 2025.
11. Anshu Gupta, [Siavash Mirarab](#) and Yatish Turakhia. Accurate, scalable, and fully automated inference of species trees from raw genome assemblies using ROADIES. *Proceedings of the National Academy of Sciences*, 122(19):e2500553122, 2025.
12. Chao Zhang, Rasmus Nielsen and [Siavash Mirarab](#). CASTER: Direct species tree inference from whole-genome alignments. *Science*, eadk9688 2025.
13. Ali Osman Berk Şapcı and [Siavash Mirarab](#). Alignment-Free Estimation of Read to Genome Distances and Its Applications. In *Research in Computational Molecular Biology*, volume 15647, 316–320. Springer Nature Switzerland, Cham, 2025. Series Title: Lecture Notes in Computer Science.
14. Puoya Tabaghi, Michael Khanzadeh, Yusu Wang and [Siavash Mirarab](#). Principal Component Analysis in Space Forms. *IEEE Transactions on Signal Processing*, 72:4428–4443, 2024.
15. Ali Osman Berk Şapcı and [Siavash Mirarab](#). Memory-bound k -mer selection for large and evolutionary diverse reference libraries. *Genome Research*, 34:1455–1467 2024.
16. Uyen Mai, Eduardo Charvel and [Siavash Mirarab](#). Expectation-Maximization enables Phylogenetic Dating under a Categorical Rate Model. *Systematic Biology*, 73(5):823–838 2024.
17. Josefin Stiller, Shaohong Feng, Al-Aabid Chowdhury et al. Complexity of avian evolution revealed by family-level genomes. *Nature*, 629:851–860 2024.
18. [Siavash Mirarab](#), Iker Rivas-González, Shaohong Feng et al. A region of suppressed recombination misleads neoavian phylogenomics. *Proceedings of the National Academy of Sciences*, 121(15):e2319506121 2024.
19. Yueyu Jiang, Daniel McDonald, Daniela Perry, Rob Knight and [Siavash Mirarab](#). Scaling DEPP phylogenetic placement to ultra-large reference trees: a tree-aware ensemble approach. *Bioinformatics*, btae361 2024.
20. Shayesteh Arasti, Puoya Tabaghi, Yasamin Tabatabaee and [Siavash Mirarab](#). Optimal Tree Metric Matching Enables Phylogenomic Branch Length Estimation. In *Research in Computational Molecular Biology*, 462–465, Cham, 2024. Springer Nature Switzerland.
21. Metin Balaban, Yueyu Jiang, Qiyun Zhu, Daniel McDonald, Rob Knight and [Siavash Mirarab](#). Generation of accurate, expandable phylogenomic trees with uDance. *Nature Biotechnology*, Online 2023.
22. Daniel McDonald, Yueyu Jiang, Metin Balaban et al. Greengenes2 unifies microbial data in a single reference tree. *Nature Biotechnology*, Online 2023.
23. Yasamin Tabatabaee, Chao Zhang, Tandy Warnow and [Siavash Mirarab](#). Phylogenomic branch length estimation using quartets. *Bioinformatics*, 39(Supplement_1):i185–i193 2023.

24. Ali Osman Berk Şapcı, Eleonora Rachtman and [Siavash Mirarab](#). CONSULT-II: Accurate taxonomic identification and profiling using locality-sensitive hashing. *Bioinformatics*, btae150 2024.
25. Shayesteh Arasti and [Siavash Mirarab](#). Median quartet tree search algorithms using optimal subtree prune and regraft. *Algorithms for Molecular Biology*, 19(1):12 2024.
26. Afif Elghraoui, [Siavash Mirarab](#), Krister M Swenson and Faramarz Valafar. Evaluating impacts of syntenic block detection strategies on rearrangement phylogeny using Mycobacterium tuberculosis isolates. *Bioinformatics*, 39(1):btad024 2023.
27. Chao Zhang, Andrey V. Bzikadze, Yana Safonova and [Siavash Mirarab](#). A scalable model for simulating multi-round antibody evolution and benchmarking of clonal tree reconstruction methods. *Frontiers in Immunology*, 13:1014439, 2022.
28. Chao Zhang and [Siavash Mirarab](#). Weighting by Gene Tree Uncertainty Improves Accuracy of Quartet-based Species Trees. *Molecular Biology and Evolution*, 39(12):msac215 2022.
29. Chao Zhang and [Siavash Mirarab](#). ASTRAL-Pro 2: ultrafast species tree reconstruction from multi-copy gene family trees. *Bioinformatics*, 38(21):4949–4950 2022.
30. Metin Balaban, Nishat A Bristy, Ahnaf Faisal, Md. Shamsuzzoha Bayzid and [Siavash Mirarab](#). Genome-wide alignment-free phylogenetic distance estimation under a no strand-bias model. *Bioinformatics Advances*, 2(1):2021.11.10.468111, 2022.
31. Maryam Rabiee and [Siavash Mirarab](#). QuCo: quartet-based co-estimation of species trees and gene trees. *Bioinformatics*, 38(Supplement_1):i413–i421 2022.
32. Yueyu Jiang, Puoya Tabaghi and [Siavash Mirarab](#). Learning Hyperbolic Embedding for Phylogenetic Tree Placement and Updates. *Biology*, 11(9):1256, 2022.
33. Navid Bin Hasan, Metin Balaban, Avijit Biswas, Md. Shamsuzzoha Bayzid and [Siavash Mirarab](#). Distance-Based Phylogenetic Placement with Statistical Support. *Biology*, 11(8):1212 2022. ISBN: 2079-7737.
34. Eleonora Rachtman, Shahab Sarmashghi, Vineet Bafna and [Siavash Mirarab](#). Quantifying the uncertainty of assembly-free genome-wide distance estimates and phylogenetic relationships using subsampling. *Cell Systems*, 13(10):817–829.e3 2022.
35. Yueyu Jiang, Metin Balaban, Qiyun Zhu and [Siavash Mirarab](#). DEPP: Deep Learning Enables Extending Species Trees using Single Genes. *Systematic Biology*, 72(1):17–34 2022.
36. Uyen Mai and [Siavash Mirarab](#). Completing gene trees without species trees in sub-quadratic time. *Bioinformatics*, 38(6):1532–1541 2022.
37. Metin Balaban, Yueyu Jiang, Daniel Roush, Qiyun Zhu and [Siavash Mirarab](#). Fast and accurate distance-based phylogenetic placement using divide and conquer. *Molecular Ecology Resources*, 22(3):1213–1227 2022.
38. Shahab Sarmashghi, Metin Balaban, Eleonora Rachtman, Behrouz Touri, [Siavash Mirarab](#) and Vineet Bafna. Estimating repeat spectra and genome length from low-coverage genome skims with RESPECT. *PLOS Computational Biology*, 17(11):e1009449 2021.
39. [Siavash Mirarab](#), Luay Nakhleh and Tandy Warnow. Multispecies Coalescent: Theory and Applications in Phylogenetics. *Annual Review of Ecology, Evolution, and Systematics*, 52(1):247–268 2021.
40. Eleonora Rachtman, Vineet Bafna and [Siavash Mirarab](#). CONSULT: accurate contamination removal using locality-sensitive hashing. *NAR Genomics and Bioinformatics*, 3(3):10.1101/2021.03.18.436035 2021.
41. Chao Zhang, Yiming Zhao, Edward Louis Braun and [Siavash Mirarab](#). TAPER: Pinpointing errors in multiple sequence alignments despite varying rates of evolution. *Methods in Ecology and Evolution*, 12(11):2145–2158 2021.
42. Niema Moshiri, Davey M Smith and [Siavash Mirarab](#). HIV Care Prioritization Using Phylogenetic Branch Length. *JAIDS Journal of Acquired Immune Deficiency Syndromes*, 86(5):626–637 2021.
43. Maryam Rabiee and [Siavash Mirarab](#). SODA: multi-locus species delimitation using quartet frequencies. *Bioinformatics*, 36(24):5623–5631 2021.
44. Shaohong Feng, Josefin Stiller, Yuan Deng et al. Dense sampling of bird diversity increases power of comparative genomics. *Nature*, 587(7833):252–257 2020.
45. Uyen Mai and [Siavash Mirarab](#). Log Transformation Improves Dating of Phylogenies. *Molecular Biology and Evolution*, 38(3):1151–1167 2021.
46. Chao Zhang, Celine Scornavacca, Erin K Molloy and [Siavash Mirarab](#). ASTRAL-Pro: Quartet-Based Species-Tree Inference despite Paralogy. *Molecular Biology and Evolution*, 37(11):3292–3307 2020.

47. Metin Balaban and [Siavash Mirarab](#). Phylogenetic double placement of mixed samples. *Bioinformatics*, 36(Supplement_1):i335–i343 2020.
48. Ekin Tilic, Erfan Sayyari, Josefin Stiller, [Siavash Mirarab](#) and Greg W Rouse. More is needed—Thousands of loci are required to elucidate the relationships of the ‘flowers of the sea’ (Sabellida, Annelida). *Molecular Phylogenetics and Evolution*, 151:106892 2020.
49. Francesco Asnicar, Andrew Maltez Thomas, Francesco Beghini et al. Precise phylogenetic analysis of microbial isolates and genomes from metagenomes using PhyloPhlAn 3.0. *Nature Communications*, 11(1):2500 2020. ISBN: 0348556977.
50. Eleonora Rachtman, Metin Balaban, Vineet Bafna and [Siavash Mirarab](#). The impact of contaminants on the accuracy of genome skimming and the effectiveness of exclusion read filters. *Molecular Ecology Resources*, 20(3):1755–0998.13135 2020.
51. Metin Balaban, Shahab Sarmashghi and [Siavash Mirarab](#). APPLES: Scalable Distance-Based Phylogenetic Placement with or without Alignments. *Systematic Biology*, 69(3):566–578 2020.
52. Maryam Rabiee and [Siavash Mirarab](#). INSTRAL: Discordance-Aware Phylogenetic Placement Using Quartet Scores. *Systematic Biology*, 69(2):384–391 2020.
53. Maryam Rabiee and [Siavash Mirarab](#). Forcing external constraints on tree inference using ASTRAL. *BMC Genomics*, 21(S2):218 2020.
54. Qiyun Zhu, Uyen Mai, Wayne Pfeiffer et al. Phylogenomics of 10,575 genomes reveals evolutionary proximity between domains Bacteria and Archaea. *Nature Communications*, 10(1):5477 2019.
55. James H Leebens-Mack, Michael S Barker, Eric J Carpenter et al. One thousand plant transcriptomes and the phylogenomics of green plants. *Nature*, 574(7780):679–685 2019.
56. Erfan Sayyari, Ban Kawas and [Siavash Mirarab](#). TADA: phylogenetic augmentation of microbiome samples enhances phenotype classification. *Bioinformatics*, 35(14):i31–i40 2019.
57. Lei Chen, Qiang Qiu, Yu Jiang et al. Large-scale ruminant genome sequencing provides insights into their evolution and distinct traits. *Science*, 364(6446):eaav6202 2019.
58. Shahab Sarmashghi, Kristine Bohmann, M. Thomas P. Gilbert, Vineet Bafna and [Siavash Mirarab](#). Skmer: assembly-free and alignment-free sample identification using genome skims. *Genome Biology*, 20(1):34 2019.
59. John Yin, Chao Zhang and [Siavash Mirarab](#). ASTRAL-MP: scaling ASTRAL to very large datasets using randomization and parallelization. *Bioinformatics*, 35(20):3961–3969 2019.
60. Niema Moshiri, Manon Ragonnet-Cronin, Joel O Wertheim and [Siavash Mirarab](#). FAVITES: simultaneous simulation of transmission networks, phylogenetic trees and sequences. *Bioinformatics*, 35(11):1852–1861 2019.
61. Peter Houde, Edward L Braun, Nitish Narula, Uriel Minjares and [Siavash Mirarab](#). Phylogenetic Signal of Indels and the Neoavian Radiation. *Diversity*, 11(7):108 2019.
62. Maryam Rabiee, Erfan Sayyari and [Siavash Mirarab](#). Multi-allele species reconstruction using ASTRAL. *Molecular Phylogenetics and Evolution*, 130:286–296 2019.
63. Shyam Gopalakrishnan, Mikkel-Holger S. Sinding, Jazmín Ramos-Madriral et al. Interspecific Gene Flow Shaped the Evolution of the Genus *Canis*. *Current Biology*, 28(21):3441–3449.e5 2018.
64. Daniel McDonald, Embriette Hyde, Justine W. Debelius et al. American Gut: an Open Platform for Citizen Science Microbiome Research. *mSystems*, 3(3) 2018.
65. Wolf L. Eiserhardt, Alexandre Antonelli, Dominic J. Bennett et al. A roadmap for global synthesis of the plant tree of life. *American Journal of Botany*, 105(3):614–622 2018.
66. Stefan Janssen, Daniel McDonald, Antonio Gonzalez et al. Phylogenetic Placement of Exact Amplicon Sequences Improves Associations with Clinical Information. *mSystems*, 3(3):e00021–18 2018.
67. Chao Zhang, Maryam Rabiee, Erfan Sayyari and [Siavash Mirarab](#). ASTRAL-III: polynomial time species tree reconstruction from partially resolved gene trees. *BMC Bioinformatics*, 19(S6):153 2018.
68. Uyen Mai and [Siavash Mirarab](#). TreeShrink: fast and accurate detection of outlier long branches in collections of phylogenetic trees. *BMC Genomics*, 19(S5):272 2018. ISBN: 9783319679785.
69. Erfan Sayyari and [Siavash Mirarab](#). Testing for Polytomies in Phylogenetic Species Trees Using Quartet Frequencies. *Genes*, 9(3):132 2018.
70. Erfan Sayyari, James B. Whitfield and [Siavash Mirarab](#). DiscoVista: Interpretable visualizations of gene tree discordance. *Molecular Phylogenetics and Evolution*, 122:110–115 2018. arXiv: 1709.09305.
71. Ali Akbari, Joseph J. Vitti, Arya Iranmehr et al. Identifying the favored mutation in a positive selective sweep. *Nature Methods*, 15(4):279–282 2018.

72. Niema Moshiri and [Siavash Mirarab](#). A Two-State Model of Tree Evolution and Its Applications to Alu Retrotransposition. *Systematic Biology*, 67(3):475–489 2018.
73. Erfan Sayyari, James B Whitfield and [Siavash Mirarab](#). Fragmentary Gene Sequences Negatively Impact Gene Tree and Species Tree Reconstruction. *Molecular Biology and Evolution*, 34(12):3279–3291 2017.
74. Uyen Mai, Erfan Sayyari and [Siavash Mirarab](#). Minimum variance rooting of phylogenetic trees and implications for species tree reconstruction. *PLOS ONE*, 12(8):e0182238 2017. ISBN: 1111111111.
75. [Siavash Mirarab](#). Phylogenomics: Constrained gene tree inference. *Nature Ecology & Evolution*, 1(2):0056 2017.
76. Erfan Sayyari and [Siavash Mirarab](#). Anchoring quartet-based phylogenetic distances and applications to species tree reconstruction. *BMC Genomics*, 17(S10):101–113 2016.
77. Nam-phuong N.-P. Nguyen, Michael Nute, [Siavash Mirarab](#) and Tandy Warnow. HIPPI: highly accurate protein family classification with ensembles of HMMs. *BMC Genomics*, 17(S10):765 2016.
78. James E Tarver, Mario dos Reis, [Siavash Mirarab](#) et al. The Interrelationships of Placental Mammals and the Limits of Phylogenetic Inference. *Genome Biology and Evolution*, 8(2):330–344 2016.
79. [Siavash Mirarab](#), Md. Shamsuzzoha Bayzid, Bastien Boussau and Tandy Warnow. Response to Comment on “Statistical binning enables an accurate coalescent-based estimation of the avian tree”. *Science*, 350(6257):171 2015.
80. Joel Cracraft, Peter Houde, Simon Y W Ho et al. Response to Comment on “Whole-genome analyses resolve early branches in the tree of life of modern birds”. *Science*, 349(6255):1460 2015.
81. Ruth Davidson, Pranjal Vachaspati, [Siavash Mirarab](#) and Tandy Warnow. Phylogenomic species tree estimation in the presence of incomplete lineage sorting and horizontal gene transfer. *BMC Genomics*, 16(Suppl 10):S1, 2015.
82. Jed Chou, Ashu Gupta, Shashank Yaduvanshi et al. A comparative study of SVDquartets and other coalescent-based species tree estimation methods. *BMC Genomics*, 16(Suppl 10):S2, 2015.
83. [Siavash Mirarab](#) and Tandy Warnow. ASTRAL-II: Coalescent-based species tree estimation with many hundreds of taxa and thousands of genes. *Bioinformatics*, 31(12):i44–i52 2015.
84. Nam-phuong Nguyen, [Siavash Mirarab](#), Keerthana Kumar and Tandy Warnow. Ultra-large alignments using phylogeny-aware profiles. *Genome Biology*, 16(1):124 2015.
85. Md. Shamsuzzoha Bayzid, [Siavash Mirarab](#), Bastien Boussau and Tandy Warnow. Weighted Statistical Binning: Enabling Statistically Consistent Genome-Scale Phylogenetic Analyses. *PLoS ONE*, 10(6):e0129183 2015.
86. Erich D Jarvis, [Siavash Mirarab](#), Andre J Aberer et al. Phylogenomic analyses data of the avian phylogenomics project. *GigaScience*, 4(1):1–9 2015.
87. [Siavash Mirarab](#), Nam Nguyen, Sheng Guo, Li-San Wang, Junhyong Kim and Tandy Warnow. PASTA: Ultra-Large Multiple Sequence Alignment for Nucleotide and Amino-Acid Sequences. *Journal of computational biology*, 22(5):377–86 2015.
88. E. D. Jarvis, [Siavash Mirarab](#), A. J. Aberer et al. Whole-genome analyses resolve early branches in the tree of life of modern birds. *Science*, 346(6215):1320–1331 2014.
89. [Siavash Mirarab](#), M. S. Bayzid, Bastien Boussau and Tandy Warnow. Statistical binning enables an accurate coalescent-based estimation of the avian tree. *Science*, 346(6215):1250463–1250463 2014.
90. Naim Matasci, Ling-Hong L.-H. Hung, Zhixiang Yan et al. Data access for the 1,000 Plants (1KP) project. *GigaScience*, 3(1):17, 2014.
91. Norman J. Wickett, [Siavash Mirarab](#), Nam Nguyen et al. Phylotranscriptomic analysis of the origin and early diversification of land plants. *Proceedings of the National Academy of Sciences*, 111(45):E4859–4868 2014. arXiv: 1301.5277v2 ISBN: 0027-8424.
92. [Siavash Mirarab](#), Md Shamsuzzoha Bayzid and Tandy Warnow. Evaluating Summary Methods for Multilocus Species Tree Estimation in the Presence of Incomplete Lineage Sorting. *Systematic Biology*, 65(3):366–380 2016. ISBN: 1063-5157.
93. [Siavash Mirarab](#), Rezwana Reaz, Md. Shamsuzzoha Bayzid, Théo Zimmermann, M. S. Swenson and Tandy Warnow. ASTRAL: genome-scale coalescent-based species tree estimation. *Bioinformatics*, 30(17):i541–i548 2014.
94. Nam-phuong Nguyen, [Siavash Mirarab](#), Bo Liu, Mihai Pop and Tandy Warnow. TIPP: taxonomic identification and phylogenetic profiling. *Bioinformatics*, 30(24):3548–3555 2014.

95. Arlin Stoltzfus, Hilmar Lapp, Naim Matasci et al. Phylotastic! Making tree-of-life knowledge accessible, reusable and convenient. *BMC bioinformatics*, 14(1):158, 2013. ISBN: 1471-2105 (Electronic)\r1471-2105 (Linking).
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97. Siavash Mirarab, Nam Nguyen and Tandy Warnow. SEPP: SATé-Enabled Phylogenetic Placement. In *Pacific Symposium On Biocomputing*, 247–58. WORLD SCIENTIFIC 2012.
98. Siavash Mirarab, Soroush Akhlaghi and Ladan Tahvildari. Size-constrained regression test case selection using multicriteria optimization. *Software Engineering, IEEE Transactions on*, 38(4):936–956, 2012.
99. Siavash Mirarab and Tandy Warnow. FastSP: linear time calculation of alignment accuracy. *Bioinformatics*, 27(23):3250–8, 2011. ISBN: 1367-4811; 1367-4803.
100. Hyunsook Do, Siavash Mirarab, Ladan Tahvildari and Gregg Rothermel. The effects of time constraints on test case prioritization: A series of controlled experiments. *Software Engineering, IEEE Transactions on*, 36(5):593–617, 2010.
101. Siavash Mirarab and Ladan Tahvildari. An empirical study on bayesian network-based approach for test case prioritization. In *Software Testing, Verification, and Validation, 2008 1st International Conference on*, 278–287. IEEE, 2008.
102. Siavash Mirarab, Afshar Ganjali, Ladan Tahvildari, Shimin Li, Weining Liu and Mike Morrissey. A requirement-based software testing framework: An industrial practice. In *Software Maintenance, 2008. ICSM 2008. IEEE International Conference on*, 452–455. IEEE, 2008.
103. Hyunsook Do, Siavash Mirarab, Ladan Tahvildari and Gregg Rothermel. An empirical study of the effect of time constraints on the cost-benefits of regression testing. In *Proceedings of the 16th ACM SIGSOFT International Symposium on Foundations of software engineering*, 71–82. ACM, 2008.
104. Mehdi Amoui, Mazeiar Salehie, Siavash Mirarab and Ladan Tahvildari. Adaptive action selection in autonomic software using reinforcement learning. In *Autonomic and Autonomous Systems, 2008. ICAS 2008. Fourth International Conference on*, 175–181. IEEE, 2008.
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SELECTED
SOFTWARE
(PUBLICLY
AVAILABLE)

Please refer to <http://eceweb.ucsd.edu/~smirarab/software.html>

ASTRAL (<https://github.com/smirarab/ASTRAL>) and

ASTER (<https://github.com/chaoszhang/ASTER>)

Coalescent-based species tree estimation from gene trees and alignments

PASTA (<https://github.com/smirarab/pasta>)

Multiple sequence alignment of ultra-large datasets (up to a million sequences)

SEPP,TIPP,UPP (<https://github.com/smirarab/sepp>)

Phylogenetic placement and taxonomic identification of metagenomic reads

TreeShrink (<https://github.com/uym2/TreeShrink>)

Finding and removing spuriously long branches

TreeCluster (<https://github.com/niemasd/TreeCluster>)

Clustering leaves of a phylogeny in linear time

DEPP (<https://github.com/yueyujang/DEPP>)

Deep learning for phylogenetic placement

krepp (<https://github.com/bo1929/krepp>)

Read mapping to many genomes using k -mers.